

Homework 3

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要求: Generate non-homogeneous Poisson process by transformation of the time scale.

假设 intensity function: $\rho(t) = 1 + 0.2t$, 时间区间为 $t \in [0, 50]$ 。

求累积强度函数 $\mu(t)$ 及其反函数 $\mu^{-1}(s)$ 。

- $\mu(t)$:

$$s = \mu(t) = \int_0^t \rho(u) du = \int_0^t (1 + 0.2u) du = t + 0.1t^2$$

当 $t = 50$ 时, s 的最大值为 $S_{max} = 50 + 0.1 \times 50^2 = 300$ 。

- $\mu^{-1}(s)$:

解方程 $0.1t^2 + t - s = 0$:

$$t = \mu^{-1}(s) = \frac{-1 + \sqrt{1 - 4(0.1)(-s)}}{0.2} = \frac{\sqrt{1 + 0.4s} - 1}{0.2}$$

先在 $[0, 300]$ 区间内生成一个强度为 $\rho(s)^* = 1$ 的齐次泊松过程, 然后将生成的事件时间点 s_i 代入反函数 $\mu^{-1}(s)$, 映射回真实的非齐次泊松过程时间点 t_i 。

```
set.seed(42)
t_max <- 50
s_max <- t_max + 0.1 * t_max^2

# mu^{-1}(s)
inv_mu <- function(s)
{
  return ((sqrt(1 + 0.4 * s) - 1) / 0.2)
}
```

```

# 生成齐次泊松过程
n <- 500
arrival_t <- rexp(n, rate = 1) # 时间间隔服从指数分布
s <- cumsum(arrival_t)
s_events <- s[s <= s_max] # 截断超出 s_max 的事件
t_events <- inv_mu(s_events) # 映射回非齐次泊松过程
cat(" 生成的非齐次泊松过程事件时间点:\n")

```

生成的非齐次泊松过程事件时间点:

```
print(round(t_events, 3))
```

```

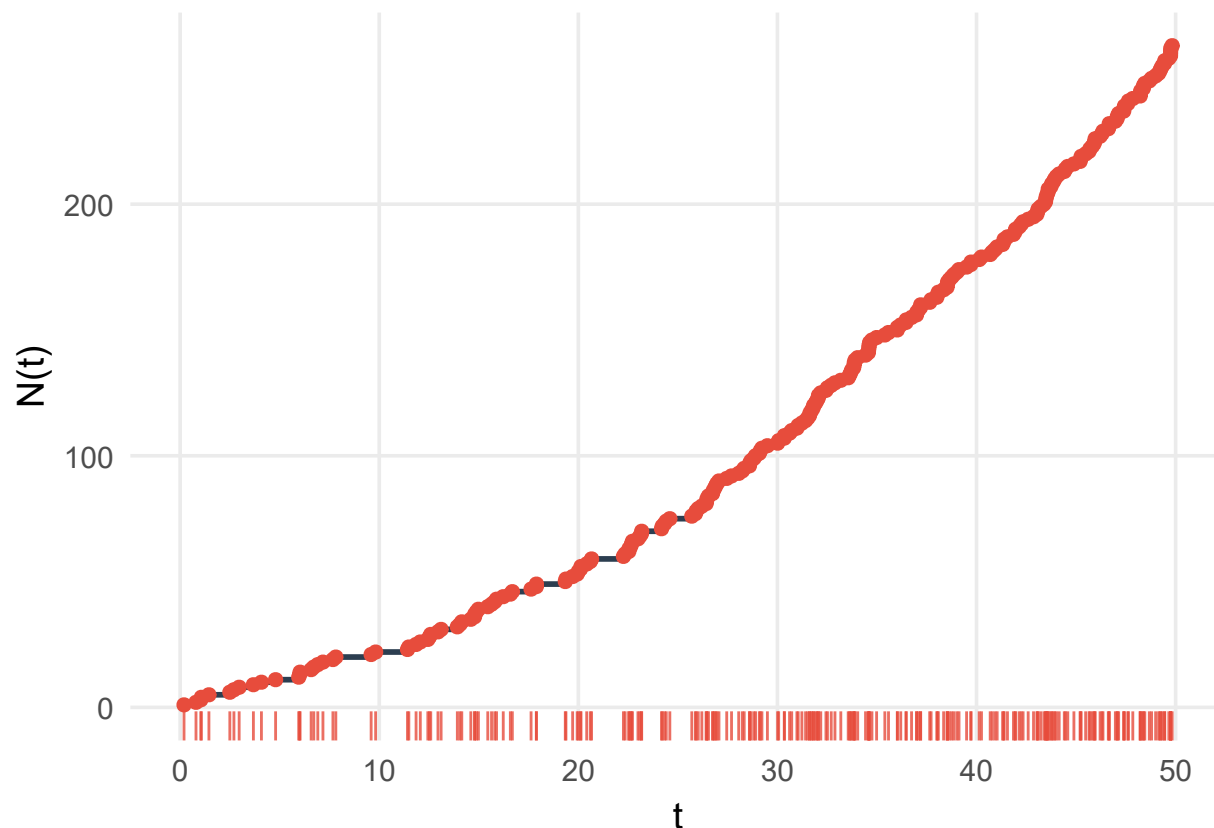
##   [1]  0.195  0.796  1.035  1.067  1.445  2.495  2.702  2.964  3.680  4.082
##  [11]  4.794  5.955  5.999  6.025  6.580  6.713  6.914  7.172  7.673  7.819
##  [21]  9.593  9.819 11.418 11.486 11.850 12.062 12.441 12.521 12.587 12.949
##  [31] 13.107 13.922 14.053 14.146 14.599 14.787 14.794 14.882 14.981 15.451
##  [41] 15.641 15.799 15.892 16.228 16.584 16.685 17.626 17.888 17.895 19.344
##  [51] 19.378 19.714 19.941 19.999 20.109 20.135 20.421 20.600 20.665 22.264
##  [61] 22.350 22.537 22.542 22.630 22.674 22.719 22.995 23.086 23.169 23.182
##  [71] 24.177 24.215 24.330 24.415 24.596 25.699 25.880 25.916 26.023 26.203
##  [81] 26.416 26.439 26.468 26.538 26.739 26.764 26.835 26.901 26.963 27.064
##  [91] 27.453 27.695 28.054 28.235 28.331 28.592 28.607 28.673 28.819 28.901
## [101] 29.088 29.141 29.221 29.496 30.009 30.063 30.342 30.352 30.609 30.717
## [111] 30.979 31.042 31.228 31.414 31.522 31.610 31.636 31.716 31.789 31.819
## [121] 31.918 31.984 32.050 32.063 32.177 32.437 32.513 32.701 32.889 33.183
## [131] 33.554 33.617 33.677 33.724 33.827 33.850 33.860 33.899 34.032 34.425
## [141] 34.566 34.578 34.593 34.603 34.631 34.747 34.981 35.394 35.579 36.019
## [151] 36.026 36.194 36.456 36.470 36.736 36.984 36.996 37.091 37.189 37.196
## [161] 37.648 37.718 38.005 38.032 38.082 38.348 38.517 38.533 38.544 38.634
## [171] 38.753 38.864 39.013 39.120 39.501 39.712 39.725 40.133 40.249 40.686
## [181] 40.790 40.925 41.036 41.309 41.364 41.387 41.553 41.854 41.933 41.962
## [191] 42.138 42.241 42.338 42.594 42.863 43.029 43.061 43.092 43.223 43.381
## [201] 43.465 43.478 43.504 43.550 43.601 43.601 43.734 43.770 43.863 43.928
## [211] 44.016 44.151 44.408 44.469 44.583 44.891 45.200 45.226 45.256 45.480
## [221] 45.649 45.696 45.811 45.898 45.938 45.958 46.205 46.301 46.351 46.618
## [231] 46.659 46.666 46.973 47.058 47.077 47.138 47.391 47.397 47.432 47.596
## [241] 47.632 47.849 48.219 48.221 48.246 48.374 48.381 48.452 48.678 48.794
## [251] 48.996 49.145 49.212 49.263 49.331 49.446 49.454 49.659 49.744 49.747
## [261] 49.749 49.758 49.826

```

```

# 画出非齐次泊松过程事件的时间点
df <- data.frame(
  t = c(0, t_events),
  n = c(0, 1:length(t_events))
)
rug_t <- data.frame(t = t_events)
p1 <- ggplot() +
  geom_step(data = df, aes(x = t, y = n), color = "#2c3e50", linewidth = 1, direction = "hv") +
  geom_point(data = df[-1, ], aes(x = t, y = n), color = "#e74c3c", size = 2) +
  geom_rug(data = rug_t, aes(x = t), color = "#e74c3c", sides = "b", alpha = 0.8, length = unit(0.
  scale_x_continuous(
    limits = c(0, t_max)) +
  labs(
    x = "t",
    y = "N(t)"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    panel.grid.minor = element_blank()
  )
print(p1)

```



横坐标为时间 t ，纵坐标为事件发生的数量 $N(t)$ ，横坐标下的红线代表事件发生的时间点。观察到随着时间的增加，事件发生的频率也在增加，这与强度函数 $\rho(t) = 1 + 0.2t$ 的线性增长趋势是一致的。

```
dens <- density(t_events, from = 0, to = t_max, bw = "nrd0")
fit <- data.frame(
  t = dens$x,
  rho = dens$y * length(t_events)
)

p2 <- ggplot(data = data.frame(t = c(0, t_max)), aes(x = t)) +
  stat_function(fun = function(t) 1 + 0.2 * t, aes(color = "intensity"), linewidth = 1.2) +
  geom_line(data = fit, aes(x = t, y = rho, color = "KDE"), linetype = "dashed", linewidth = 1) +
  geom_rug(data = rug_t, aes(x = t), color = "#e74c3c", sides = "b", alpha = 0.8, length = unit(0.1))

scale_color_manual(name = NULL, values = c("intensity" = "#27ae60", "KDE" = "blue")) +
labs(
  x = "t",
  y = expression(rho(t))
)
```

```

) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.minor = element_blank(),
  legend.position = c(0.9, 0.95),
  legend.background = element_rect(fill = "white", color = "lightgrey")
)

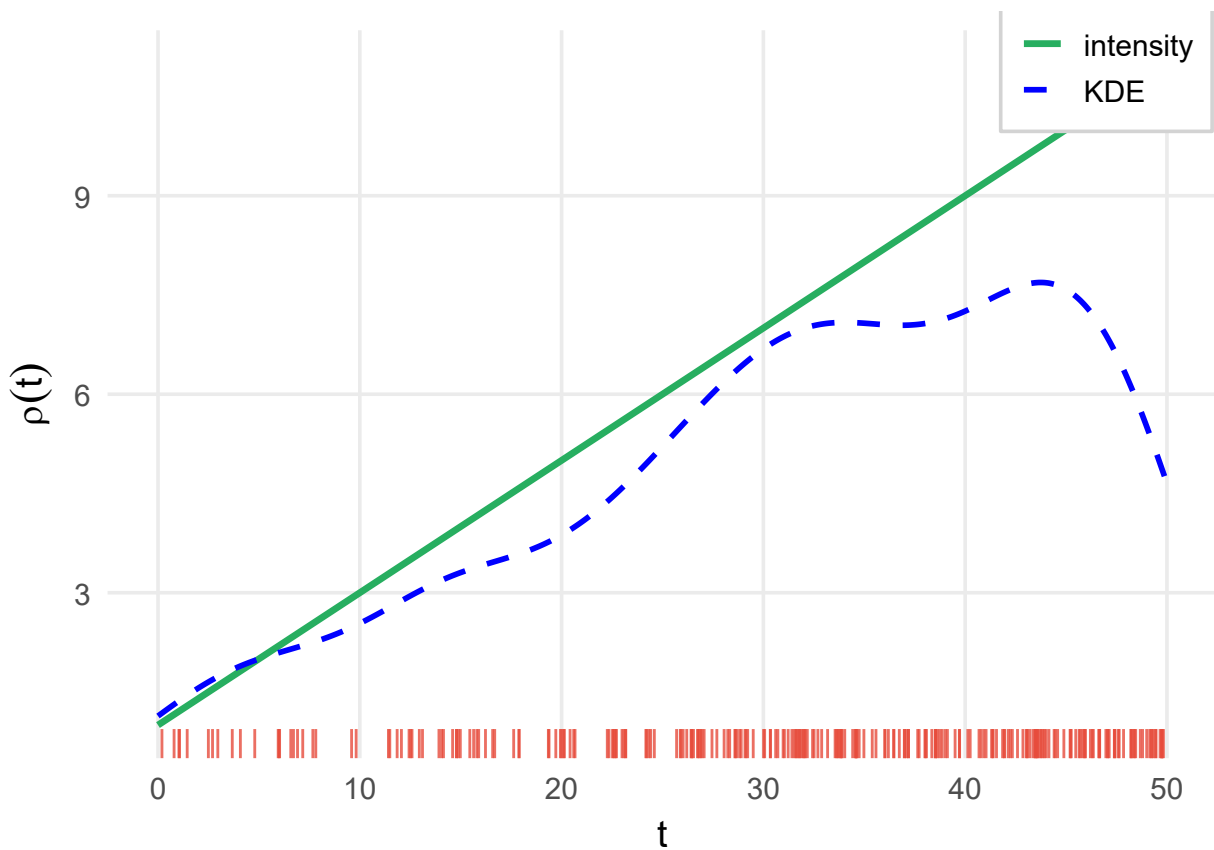
```

```

## Warning: A numeric `legend.position` argument in `theme()` was deprecated in ggplot2
## 3.5.0.
## i Please use the `legend.position.inside` argument of `theme()` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```

```
print(p2)
```



对于单一的某次模拟，用 KDE 估计事件的密度分布，并与真实的 intensity function 进行比较，可以看到前半段的估计较为接近。后半段 KDE 突然下降的原因在于 $t = 50$ 的右边被截断了，没有任何事件发生，影响了其对密度的估计。

用蒙特卡洛方法估计非齐次泊松过程的期望事件数 (cumulative intensity function) ，取 500 个时间点，重复 10 次模拟，计算每个时间点的事件数的平均值，与真实的 cumulative intensity function ($t + 0.1t^2$) 进行比较。可以看出，仅仅经过 10 次蒙特卡洛模拟，拟合的 cumulative intensity function 已经和真实的 cumulative intensity function 非常接近了，说明这种转变时间尺度生成非齐次泊松过程的方法是较为有效的。

```
rep <- 10
grid <- seq(0, t_max, length.out = 500)
Nt <- matrix(0, nrow = rep, ncol = length(grid))

# 10 次蒙特卡洛模拟 ---
for (i in 1:rep) {
  arr_t <- rexp(500, rate = 1)
  ss <- cumsum(arr_t)
  ss_events <- ss[ss <= s_max]
  tt_events <- inv_mu(ss_events)
  Nt[i, ] <- sapply(grid, function(t) sum(tt_events <= t)) # 时间网格点上的事件数
}

observed <- colMeans(Nt)
theory <- grid + 0.1 * grid^2

df_m <- data.frame(
  t = rep(grid, 2),
  Value = c(observed, theory),
  Type = rep(c("observed", "theory"), each = length(grid))
)

p3 <- ggplot(df_m, aes(x = t, y = Value, color = Type, linetype = Type)) +
  geom_line(linewidth = 1.2) +
  scale_color_manual(values = c("observed" = "#2980b9", "theory" = "#e74c3c")) +
  scale_linetype_manual(values = c("observed" = "solid", "theory" = "dashed")) +
  labs(
    x = "t",
    y = expression(E[N(t)])
  )
```

```

) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.minor = element_blank(),
  legend.position = "inside",
  legend.position.inside = c(0.2, 0.8),
  legend.background = element_rect(fill = "white", color = "lightgrey")
)
print(p3)

```

